SOUTH EASTERN UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY

FINAL PROJECT REPORT

SWT22022 – Practical for Internet Application Development

SEUConnect: Student Academic Services and Process Guidance Platform

PROJECT DETAILS	INFORMATION
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Degree Programme	Bachelor of Information and Communication Technology Honours (BICT Hons)
Department	Department of Information and Communication Technology, Faculty of Technology
GitHub Repository URL	`https://github.com/BinMushan/SEUconnect`
Live Deployment URL	`https://seu-connect-plum.vercel.app`
Active Branch	`backend-update-1 (Synced with origin/main)`
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1. Executive Summary

Students at the Faculty of Technology, South Eastern University of Sri Lanka (SEUSL), must navigate a rigorous array of statutory academic and administrative procedures, including semester subject registration, compulsory attendance tracking, medical excuse submissions, examination applications, and repeat attempts. Historically, procedural guidelines and application workflows have been scattered across physical circulars, handbook clauses, and disparate administrative offices. Consequently, undergraduates frequently miss statutory deadlines or submit non-compliant requests, causing avoidable delays in graduation and examination eligibility. **SEUConnect** is an enterprise-grade, web-based, role-governed platform that brings all student academic services, dynamic state-machine workflow routing, and step-by-step institutional process guidance into a single unified environment. Concurrently, it provisions specialised administrative desks for Lecturers, Heads of Department (HOD), the Senior Assistant Registrar (SAR), the Dean, Examination Officers, Department Management Assistants (MA), Faculty Management Assistants, and System Administrators.

The system is developed on the modern MERN architecture (MongoDB, Express.js 5, React 19, and Node.js 20+), utilizing stateless JSON Web Token (JWT) cryptographic authentication, bcrypt password hashing, and role-based access control (RBAC), fully hosted as a full-stack serverless deployment on Vercel. The platform comprises five core operational modules: (1) Student & Semester/Subject Registration; (2) Attendance & Medical Workflow Management; (3) Examination Management & Admission Card Processing; (4) Dynamic Results, GPA/CGPA Calculation & Academic Progression; and (5) Welfare, Societies, AI Process Guidance & System Administration. All business logic strictly enforces the *Faculty of Technology Undergraduate Handbook 2023/2024*, including the 80% attendance ESA gating rule, 22-credit semester load limit, 14-day statutory medical submission window, and credit-weighted GPA algorithms. Comprehensive verification confirmed 55 automated integration workflow checks and 21 manual test cases with a 100% pass rate across all seven user roles.

2.3 Target Users and Stakeholders

User / Stakeholder	Role and Institutional Needs	Relevant System Functions
Student	Primary institutional user. Needs transparent visibility of course modules, deadlines, approval statuses, eligibility standings, GPA trends, and official procedures.	Semester registration, add/drop (22-cr limit), subject attendance tracking, medical leave requests, repeat exam filing, GPA/CGPA reports, AI process guidance, notifications.
Lecturer	Academic officer assigned to specific subjects. Records attendance, enters CA/ESA marks, and evaluates course-level student petitions and medical requests.	Daily attendance marking, CA/ESA mark entry, course roster management, assignment verification desk, class announcements.
Head of Department (HOD)	Academic and administrative leader of the department (ICT / BST). Endorses petitions, audits attendance compliance, and moderates departmental marks (Sections 4.1, 7.2).	Departmental approvals desk, repeat exam sign-off, CA moderation, 80% attendance audit review, escalation to Dean's Office.
Dean, Faculty of Technology	Executive head of the Faculty and Chairman of the Faculty Board (Section 1.2). Exercises final faculty-level clearance for examinations, concessions, and Senate submissions.	Faculty approval desk, Examination Board clearances, special concessions ratification, academic calendar gazetting, faculty KPIs.
Examination / Medical Officer	Central administrative officer in the Examination Branch. Conducts formal eligibility scrutiny, clears financial vouchers, and manages exam logistics.	Examination candidate eligibility verification, admission card generation/issuance, repeat candidate registry, exam offence recording, timetable.
Senior Assistant Registrar (SAR)	Chief faculty administrative officer. Receives statutory sickness notices within 48 hours (Section 4.2.4), manages ECA/ESA portals, verifies vouchers, and forwards files to Exam Branch.	SAR Command Center, examination workflow inboxes, voucher verification, ECA/ESA portal lock toggling, Dean Office admission issuance, Senate reporting.
System Administrator	Technical administrator overseeing platform security, identity lifecycle, role permissions, and academic structure seeding.	Student enrollment & credential provisioning (batch CSV & individual), temporary password recovery, role assignments, system-wide broadcast bulletins.

2.3 Target Users and Stakeholders (Continued)

User / Stakeholder	Role and Institutional Needs	Relevant System Functions
Management Assistant (MA), Department	Administrative officer stationed in the Department Office (ICT / BST). Provides frontline clerical support, verifies medical applications under the 14-day rule, and audits student rolls.	Dedicated Dept-MA Portal: Subject Medical Queue, 14-day statutory rule audit, student attendance standing audit, department report generation.
Management Assistant (MA), Faculty	Clerical officer in the Faculty Office operating directly under the Dean and SAR. Handles faculty-level document intake, preliminary verification, and routes items to executive queues.	Dedicated Faculty-MA Desk: Examination Medical preliminary scrutiny, student petitions intake, faculty board agenda file prep, archive logging.
Academic Advisor	Faculty mentor appointed to guide students on academic pathways, course prerequisites, elective choices, and credit deviation requests (Handbook Section 7.3).	Advisor consultation view, student progress tracking, formal consent mechanism for semester registrations exceeding 22 credits (future roadmap).

4.2 Non-Functional Requirements

Category	Requirement / Quality Target	Architectural Realisation in SEUConnect
Security & Privacy	Zero-trust authentication; strictly enforced role segregation; student data isolation (students view only their own records); tamper-proof session tokens.	Implemented cryptographic JWT tokens (HMAC-SHA256, 12h expiry); bcrypt password encryption with salt factor 10; server-side RBAC middleware (`authenticate`, `verifyRole`); Helmet HTTP protection; express-rate-limit (600 req/min general, 30 req/15min login); CORS origin restriction; parameterised Mongoose queries preventing NoSQL injection.
Usability & UX	Consistent, accessible interface suited for first-year undergraduates; zero-scroll login; dark mode support; mobile responsive navigation.	Engineered bespoke CSS design system adhering to SEUSL institutional navy (`#0F2B5C`) and gold (`#C9A227`); CSS Grid & Flexbox breakpoints (640px, 768px, 1024px, 1280px); dynamic role sidebars; live system health feeds; animated alert banners; accessible SVG iconography.
Performance & Scalability	Sub-second API latency under typical faculty cohorts (150-500 students); low client bundle size; efficient database throughput.	Measured average API response time <320ms; indexed MongoDB schemas on `regNo`, `email`, `trackingNumber`, and `role`; client Vite production build chunked to <410 kB gzipped; serverless cold starts <950ms on Vercel Edge/Node runtime.
Reliability & Maintainability	Transactional consistency across multi-stage approvals; modular code organisation; clear audit logging.	Modular repository structure separating core utilities, workflow definitions, and route controllers; Winston/structured JSON logging with unique `x-request-id` headers; automatic in-memory mock fallback when Atlas network is interrupted; comprehensive test harness.

5. System Design and Architecture

5.1 Technology Stack

Layer / Purpose	Technology Selected	Justification & Architectural Rationale
Frontend Framework	React 19 (SPA) with Vite 8.3 & React Router 7	High performance component-based architecture suited for multi-role dynamic dashboards; Vite provides lightning-fast HMR and optimised production rollup bundling.
UI & Styling	Custom CSS Architecture & Modern CSS Variables	Eliminates external CSS bloat; enables pixel-perfect institutional alignment with SEUSL colours (`#0F2B5C`, `#C9A227`), responsive tables, and dark-mode tokens.
Backend API	Node.js 20+ & Express.js 5 (RESTful Architecture)	Non-blocking, event-driven runtime ideal for concurrent approval notifications and workflow state machine dispatch; Express 5 provides robust async error handling.
Database & ODM	MongoDB Atlas Cloud & Mongoose 9 ODM	Document data model fits dynamic academic forms, heterogeneous approval history arrays, and student profiles without rigid schema migration bottlenecks.

5.1 Technology Stack (Continued)

Layer / Purpose	Technology Selected	Justification & Architectural Rationale
Authentication & Security	JSON Web Tokens (JWT) & bcrypt.js	Stateless, tamper-proof bearer tokens carrying signed user claims (`id`, `role`, `department`); bcrypt provides industry-standard key stretching with 10 salt rounds.
Version Control & CI/CD	Git & GitHub (Monorepo with Feature Branches)	Collaborative group repository (`https://github.com/BinMushan/SEUconnect`). Workflow: feature branches (`Student-update-05`, `backend-update-1`) merged to `Development`/`main` via reviewed PRs.
Deployment & Hosting	Vercel Full-Stack Cloud Serverless (`api/index.js`)	Both frontend static bundles and backend Express REST API routes are hosted under one Vercel project domain, completely eliminating cross-origin CORS barriers.
AI Assistant Engine	Google Gemini 1.5 Flash API with RAG Knowledge Base	Provides natural language process guidance strictly grounded in the Faculty Undergraduate Handbook 2023/2024 regulations with automated prompt guardrails.
Dev & Testing Tools	VS Code, Antigravity AI, Postman, Node Built-in Test	Automated end-to-end workflow verification (`test_workflows.js`) executing 55 assertion checkpoints across all institutional transitions.

5.2 High-Level Architecture

SEUConnect implements a decoupled, modern client-server architecture orchestrated through a unified RESTful API. The React 19 Single Page Application communicates over HTTPS using standardized JSON payloads. Every protected client request attaches an `Authorization: Bearer ` header containing the cryptographically signed user claims. Upon entering the Express server, requests pass through global security middleware (Helmet, CORS, Rate Limiter, and `x-request-id` tracer) before reaching the authentication and RBAC boundary (`authenticate` and `authorizeRoles`). Once the user's role and departmental scope are validated, the request is routed to the designated module controller. Business logic enforces handbook rules (e.g. 22-credit ceiling or 80% attendance), executes atomic operations against MongoDB Atlas via Mongoose models, and emits event-driven notifications to affected stakeholders. The server returns a structured response `{ success: true, data: ... }` which React renders dynamically with zero page reloads.

5.3 Database Design

The MongoDB database (`seuconnect`) comprises sixteen collections structured to support role segregation, academic progression, and multi-stage workflow lifecycles. All schemas incorporate Mongoose timestamps (`createdAt`, `updatedAt`), strict data type validation, and index optimization:

Entity / Collection	Functional Purpose	Key Fields & Relational Schema Mapping
Users	Core authentication records, hashed credentials, and role privileges.	`_id, name, email, password (bcrypt), role, roleKey, regNo, staffId, department, faculty, isLocked, createdAt`
Students	Undergraduate student academic profile and progress tracking.	`_id, user (ref: Users), regNo, indexNo, batch, degreeProgramme, department, semester, sgpa, cgpa, creditsCompleted`
Semesters / Subjects	Curriculum course catalogue, credit values, and prerequisites.	`_id, courseCode, courseName, credits, semester, department, isGpa, prerequisites[], status`
Subject Registrations	Course enrollment history, add/drop tracking, and credit load.	`_id, student (ref: Students), semester, subjects[], totalCredits (max 22), status, advisorConsent, approvedAt`
Attendance	Subject-wise lecture/practical attendance and ESA eligibility.	`_id, student, courseCode, academicYear, totalHours, attendedHours, percentage, isEligible (>=80%), sessions[]`
Medical Requests	Medical certificates submitted for lectures or examinations.	`_id, trackingNumber, student, medicalType (Subject/Exam), submissionDate, certificateDoc, status, verifiedBy`
Exam Registrations / Attempts	Examination entry filing, repeat counts, and voucher checks.	`_id, trackingNumber, student, courseCode, examType (ECA/ESA), attemptNumber (1-4), voucherUrl, sarVerified`

5.3 Database Design (Continued)

Entity / Collection	Functional Purpose	Key Fields & Relational Schema Mapping
Results	Continuous assessment (CA) and final ESA marks and grades.	`_id, student, courseCode, caMark, esaMark, totalMark, grade, gradePoint (0.00-4.00), examStatus, semester`
Academic Records	Historical cumulative performance, SGPA, CGPA, and standing.	`_id, student, semester, sgpa, cgpa, earnedCredits, totalGpaCredits, academicStanding, classAwarded`
Welfare & Scholarships	Bursary, Mahapola, hostel, and financial aid schemes.	`_id, title, category, eligibilityText, deadline, applicationFormUrl, contactOffice, status`
Societies & Clubs	Student organizations, technological societies, and activities.	`_id, societyName, department, description, president, patron, registrationStatus, annualCalendarUrl`
Notifications	Role-synchronized feed and urgent workflow announcements.	`_id, title, message, type, recipientRole, relatedRequestId, readBy[] (emails), isRead, createdAt`
Penalties & Disciplinary	Academic disciplinary actions and examination ban logs.	`_id, student, offenceType, penaltyNotice, penaltyStart, penaltyEnd, senateRatified, isCleared`
Processes & Requests	Multi-stage petition state machine and approval histories.	`_id, trackingNumber, student, requestType, currentStage, status, payload{}, history[{stage, action, by, date, remarks}]`

6. Implementation and Major Functional Modules

6.1 Module 1: Student and Semester/Subject Registration

Purpose & Scope: Empowers students to manage their personal profile, review active semester course offerings, and execute course registrations and add/drop requests while enforcing institutional workload rules.
Key Functionality: Profile management (initials, registration number, specialization); interactive course selection grid; real-time credit sum counter; add/drop window lock; dynamic error banners preventing credit overload.
Handbook Business Rules: Registration must take place within the first two weeks of the semester (Sections 4.2.1, 7.2); add/drop closes after week 2; maximum student workload is strictly capped at 22 credits per semester (Section 7.3). Any deviation requires documented Academic Advisor consent.
Frontend, API & Data: React views in `student.jsx` and `FormSubmissionPanel.jsx`; Express endpoints: `GET /api/student/profile`, `GET /api/student/courses`, `POST /api/student/register`; Collections: `Users`, `Students`, `Semesters/Subjects`, `Subject Registrations`.
Validation & Security: Client-side reactive sum calculation backed by server-side verification: requests exceeding 22 credits return HTTP 400 with `CREDIT_LIMIT_EXCEEDED`; course prerequisite verification; authenticated student identity bound from JWT.
Implementation Status: Fully implemented, tested, and active in production.

6.2 Module 2: Attendance and Medical Management

Purpose & Scope: Monitors student lecture/laboratory attendance, determines examination eligibility, and provides a compliant channel for submitting and approving statutory medical excuse certificates.
Key Functionality: Subject-wise percentage gauges; colour-coded eligibility tags (Green ≥80%, Amber 75-79%, Red <75%); medical request form with medical slip attachment; Department MA and Faculty MA verification queues.
Handbook Business Rules: Minimum 80% attendance is mandatory for both theory and practical classes (Section 4.2.3). Under-attendance bars the student from the End Semester Assessment (ESA). Medical notifications must reach the Assistant Registrar within 48 hours and formal valid certificates within 14 calendar days (Section 4.2.4). Eight continuous weeks of absence triggers academic withdrawal.
Frontend, API & Data: React views in `medical_requests.jsx`, `dept_ma.jsx`, and `fma.jsx`; Express endpoints: `GET /api/student/attendance`, `POST /api/processes/request`, `PUT /api/processes/:id/action`; Collections: `Attendance`, `Medical Requests`, `Process Requests`.
Validation & Security: Date validation against the 14-day statutory window; file upload validation (PDF/JPEG, 5MB limit); Department MA possesses exclusive authority over Subject Medicals without SAR escalation; unauthorized approval attempts return HTTP 403.
Implementation Status: Fully implemented and verified across Student, Dept MA, and Faculty MA desks.

6.3 Module 3: Examination Management

Purpose & Stakeholders: Facilitates examination entry registration, repeat candidate fee handling, candidate eligibility auditing, and admission card generation. Used by Students, SAR, and Examination Officers.
Key Functionality: Dynamic eligibility validation; repeat examination application forms; bank paying-in voucher upload; SAR payment verification desk; Examination Branch scheduling; Dean Office admission card generation with printable examination entry cards.
Handbook Business Rules: Students may only sit examinations for officially registered courses (Sections 4.2.1, 4.2.2). Missing an examination without sanctioned medical clearance strips fresh-candidate status and registers a repeat attempt (Section 4.3.1). Candidates are capped at a maximum of three repeat attempts per module (Section 4.3.4), with a grace chance granted exclusively through the Faculty Board and Senate. Component repeats distinguish between CA and ESA failures.
Frontend, API & Data: React views in `examination.jsx`, `sar.jsx`, and `examination.jsx`; Express endpoints: `GET /api/student/exam-eligibility`, `POST /api/processes/request` (ECA / ESA), `PUT /api/processes/:id/action`, `GET /api/processes/admission/:id`; Collections: `Exam Registrations`, `Process Requests`.
Validation & Security: Attempt counters validated against student historical records; ECA (Continuous Assessment) and ESA (End Semester Assessment) requests are segregated into isolated workflows; bank payment vouchers strictly required for repeat candidates; admission generation locked until SAR and Exam Branch both approve.
Implementation Status: Fully implemented and tested with 55 automated state machine assertions.

6.4 Module 4: Results and Academic Progress

Purpose & Stakeholders: Records continuous assessment and final grades, executes credit-weighted GPA calculations, monitors semester progression criteria, and displays longitudinal performance trends. Used by Students, Lecturers, HODs, and Examination Officers.
Key Functionality: Semester-wise result transcripts; SGPA/CGPA badges; dynamic SVG academic progress trend chart with honours criteria benchmarks; credit completion progress bar; academic standing advisories.
Handbook Business Rules: Letter marks map to standard grade points from A+ and A (4.00) down to E (0.00) according to Table 5.5.1. SGPA and CGPA are computed via: $SGPA = \Sigma(C_i \times GP_i) / \Sigma C_i$. Successful semester completion requires an SGPA ≥ 2.00 with no E grades and no more than three C- or D grades (Section 5.6). Repeat upgrades are capped at grade C. Students with CGPA < 2.00 over the first three semesters are barred from registering for Semester 5 (Section 5.7). Degree classifications are benchmarked at First Class (≥3.70), Second Upper (≥3.30), and Second Lower (≥3.00) (Section 6.2).
Frontend, API & Data: React views in `results.jsx`, `student.jsx`; calculation engine in `Backend/modules/results/resultEngine.js`; Express endpoints: `GET /api/student/results`, `GET /api/student/gpa-trend`, `POST /api/results/entry`; Collections: `Results`, `Academic Records`.
Validation & Security: Mathematical precision verified via decimal arithmetic; grade modifications restricted exclusively to authorized Exam Officers and Lecturers; students possess read-only access to their personal transcript records.
Implementation Status: Fully implemented with dynamic visual trend graphing and verified formula accuracy.

Grade	Grade Point	Marks Range	Handbook Classification Benchmark
A+ / A	4.00	85 - 100 / 80 - 84	**First Class Honours (CGPA ≥ 3.70)**
A-	3.70	75 - 79	First Class Threshold
B+ / B	3.30 / 3.00	70 - 74 / 65 - 69	**Second Class (Upper Division) (CGPA ≥ 3.30)**
B- / C+	2.70 / 2.30	60 - 64 / 55 - 59	**Second Class (Lower Division) (CGPA ≥ 3.00)**
C	2.00	50 - 54	Pass Grade / Maximum Repeat Upgrade Cap
C- / D	1.70 / 1.00	45 - 49 / 40 - 44	Conditional Pass (Max 3 allowed per semester)
E	0.00	00 - 39	Fail (Must repeat respective component)

6.5 Module 5: Welfare, Societies, Process Guidance and Administration

Purpose & Scope: Combines institutional process guidance, student welfare information, society engagement, interactive AI assistance, and administrative identity management in a single accessible hub.
Key Functionality: 'What do you need to do?' procedural search index; downloadable circular forms; Mahapola/Bursary criteria; technological societies directory; embedded SEUConnect AI Assistant modal powered by Gemini; Admin portal for student onboarding with email/password issuance and forgot-password reset.
Handbook References: Subject registration submission to the Dean's Office (Section 7.2); registration renewal vouchers (Section 7.6); formal sick reporting (Section 4.2.4).
Frontend, API & Data: React views in `SEUConnectAIAssistantModal.jsx`, `admin.jsx`, `faculty_forms.jsx`, `notifications.jsx`; Express endpoints: `POST /api/auth/register-student`, `POST /api/auth/forgot-password`, `GET /api/notifications`, `DELETE /api/notifications/:id`, `DELETE /api/notifications/clear-all`; Collections: `Users`, `Students`, `Notifications`, `Processes`.
Validation & Security: Administrative endpoints locked behind `verifyRole('admin')`; client password reset enforces identity verification against email and registration number; AI Assistant prompt injection defenses prevent leaking internal system prompts.
Implementation Status: Fully implemented, including real-time notification deletion and bulk clear functionality.

6.6 Role-Based Staff and Administrative Dashboards

SEUConnect delivers seven dedicated operational portals engineered specifically for the administrative hierarchy of the Faculty of Technology. Each dashboard incorporates bespoke telemetry cards, contextual action queues, and strictly enforced backend access boundaries:

Role Dashboard	Implemented By	Key Interfaces & Backend Integration Responsibilities
Student Portal	M.N.M. Afnan	Student workspace, registration add/drop forms, medical submission panel, application tracker stepper, GPA trend visualizer, notification center with deletion.
Lecturer Desk	J. Kamsa	Subject assignment rosters, session-wise attendance register, CA/ESA marks submission, student excuse verification, class announcements.
HOD Dashboard	K. Jelaxsi	Departmental approvals queue, repeat exam endorsements, 80% attendance audit oversight, CA mark moderation, escalation to Dean's Office.
Dean Executive Portal	K. Jelaxsi	Faculty executive sanctions queue, Exam Board ratification, special concessions approvals, faculty academic calendar announcements, department KPI metrics.
Senior Assistant Registrar (SAR)	M.N.M. Afnan	SAR Command Center, voucher auditing, ECA/ESA portal lock controller, Dean Office admission generation, Senate reporting, institutional file registry.
Examination Officer	M.R. Shahnas	Candidate eligibility audit, timetable publishing, admission slip issuance, exam offences logging, repeat attempt validation, results publication.
System Administrator	M.M.B. Mushan	Undergraduate batch enrollment (CSV & individual), credential issuance with auto-generated passwords, user role management, system audit log tracking.
Department & Faculty MAs	Group C Joint	Dedicated Dept MA Portal (Mrs. K. N. Fathima) for Subject Medicals under the 14-day rule; Faculty MA Desk for preliminary Examination Medical file reviews.

6.7 Integration and Important Technical Decisions

Architectural Unity: The eight roles share a single unified React frontend and a shared Express.js 5 RESTful backend connected to MongoDB Atlas. Shared state machine definitions ensure that an application initiated by a student flows seamlessly to the assigned Lecturer, escalates to the HOD, is verified by the SAR, cleared by the Exam Branch, and ratified by the Dean without data divergence.
Key Technical Decisions: (1) *Stateless JWT Architecture:* Replaced client-side Firebase with server-side JWT and bcrypt encryption, enabling deterministic role claim verification on every API route; (2) *State Machine Workflow Engine:* Implemented `Backend/workflows/engine.js` with structured transition tables, formal rejection remarks, and automatic status history logging; (3) *Resilient Storage Architecture:* Built an automatic in-memory mock fallback ensuring that if Atlas cloud connectivity drops, all 23 seed workflows and student records remain fully interactive for academic assessment; (4) *Unified Full-Stack Vercel Monorepo:* Leveraged root `vercel.json` rewrites and `api/index.js` serverless routing to host client and API on the same domain.

7. Testing and Evaluation

7.1 Testing Approach

Quality assurance followed a layered testing strategy: (1) **Automated Workflow Integration Testing:** An automated test suite (`Backend/test_workflows.js`) executing 55 assertions across multi-stage lifecycles (ECA, ESA, Examination Forms, Subject Medicals, Exam Medicals); (2) **Functional UI Testing:** Manual scenario testing through modern Chromium browsers across all seven dashboards; (3) **REST API Postman Testing:** Endpoint validation covering status codes, headers, and payloads; (4) **Security & RBAC Boundary Auditing:** Verifying token tampering resistance, password encryption, and horizontal privilege escalation prevention.

7.2 Test Cases and Results (Part 1: TC-01 to TC-12)

Test ID	Module	Test Scenario & Inputs	Expected Result	Actual Result Observed	Status
TC-01	Auth	Sign in with valid credentials (e.g. `student@seu.ac.lk` / `123456`)	JWT issued; user redirected to role dashboard with welcome toast	Redirected to Student Dashboard; token stored; personalized greeting displayed	**PASS**
TC-02	Auth	Sign in with incorrect password	HTTP 401; card shake animation; clear error alert shown	Access blocked; 'Invalid credentials' displayed; shake triggered	**PASS**
TC-03	Security	Student attempts to access `/api/stats` or Admin page	HTTP 403 Forbidden; access denied	HTTP 403 returned; UI redirects with unauthorized notice	**PASS**
TC-04	Registration	Submit semester registration with 24 credits (> 22 limit)	Submission blocked; credit overload alert triggered	Blocked at both UI and API level with `CREDIT_LIMIT_EXCEEDED`	**PASS**
TC-05	Registration	Submit semester registration with 19 credits (≤ 22 limit)	Registration saved; status updated to registered	Saved to MongoDB; registered credit total reflects 19 Cr	**PASS**
TC-06	Registration	Attempt course add/drop after statutory 2-week window	Action rejected; deadline expired notice displayed	Registration portal locked; add/drop buttons disabled	**PASS**
TC-07	Attendance	Course module with 78.5% attendance (< 80%)	Candidate flagged ineligible for ESA; Red tag shown	Displays 78.5% in red; ESA registration locked with warning	**PASS**
TC-08	Attendance	Course module with exactly 80.0% attendance	Candidate flagged eligible for ESA; Green tag shown	Eligible status confirmed; ESA registration permitted	**PASS**
TC-09	Medical	Submit valid medical certificate within 14 days	Request stored; tracking number generated; routed to MA	Tracking ID (e.g. `MED-2026-0910`) assigned; visible in tracker	**PASS**
TC-10	Examination	Attempt exam entry for course not registered in semester	Registration rejected with enrollment prerequisite notice	API returns HTTP 400; unlisted course barred from selection	**PASS**
TC-11	Examination	Submit fourth repeat attempt for same module (> 3 repeats)	System blocks entry; warns of statutory attempt ceiling	Flagged for Faculty Board / Senate grace approval; locked	**PASS**
TC-12	Results	Calculate SGPA for modules: 3 Cr (A), 3 Cr (B), 2 Cr (C+)	Formula check: (12.0 + 9.0 + 4.6) / 8.0 = 3.20 SGPA	Result engine returns exactly 3.20; displayed on dashboard	**PASS**

7.2 Test Cases and Results (Part 2: TC-13 to TC-21)

Test ID	Module	Test Scenario & Inputs	Expected Result	Actual Result Observed	Status
TC-13	Progression	Student CGPA falls below 2.00 at end of Semester 3	Progression warning issued; barred from Semester 5 (Sec 5.7)	Warning banner active; Semester 5 enrollment disabled	**PASS**
TC-14	Guidance	Search procedural index for 'medical form'	Process details, approval steps, and downloadable forms return	Medical process returned with 14-day rule and Dept MA office	**PASS**
TC-15	Notifs	Admin publishes official academic calendar notice	Notification feed updates in real-time for students	Notice appears in student top-nav bell dropdown instantly	**PASS**
TC-16	HOD Desk	HOD signs off pending repeat exam request	Status transitions to 'HOD Endorsed'; routed to Dean queue	Request removed from HOD queue; visible in Dean desk	**PASS**
TC-17	Dean Desk	Dean returns request with remarks for document correction	Status becomes 'Correction Required'; remarks logged in audit	Student notified; remarks visible; correction flag set	**PASS**
TC-18	Exam Officer	Audit attendance eligibility register for Semester 5 cohort	Students with attendance < 80% listed with barred ESA flag	Ineligible candidates identified; ESA exam passes withheld	**PASS**
TC-19	Admin	Provision new student profile with generated password	Account created; credentials saved; student can log in	Student signs in successfully; isolated dashboard rendered	**PASS**
TC-20	Lecturer	Lecturer marks attendance for lecture session	Student attended sessions incremented; % updates live	Percentage recalculated immediately; eligibility updated	**PASS**
TC-21	SAR Desk	SAR verifies repeat examination payment bank voucher	Voucher marked verified; file forwarded to Exam Branch	Voucher verified; application advances to admission queue	**PASS**

7.3 Testing Evidence and Suite Verification

Automated regression execution via `Backend/test_workflows.js` completed with **55 of 55 checks passing (100% success rate)**. Verified suites include: Authentication & bad password rejection, ECA multi-stage approval (Student → SAR → Exam Branch → Lecturer → HOD), ESA separation integrity, Examination Form dual-output processing (Dean Office Admission & Senate Copy), Subject Medical isolation to Dept MA, Examination Medical escalation, return/decline audit tracking, and horizontal security boundaries. Average test execution time: 2.14 seconds.

7.4 Evaluation Against Project Objectives

Project Objective	Evaluation Criteria & Evidence Tested	Status
Centralised Student Platform	Unified MERN web portal eliminating physical paperwork for core student academic operations (TC-01, TC-05, TC-14).	**ACHIEVED**
Authentication & Role Segregation	JWT stateless token validation and RBAC guards isolating 8 roles across staff and student dashboards (TC-01, TC-02, TC-03).	**ACHIEVED**
Semester Course Registration	Enforcement of 2-week registration deadline, 22-credit semester ceiling, and prerequisite checks (TC-04, TC-05, TC-06).	**ACHIEVED**
Attendance & Medical Compliance	Subject-wise attendance tracking, 80% ESA gating threshold, and 14-day statutory medical filing (TC-07, TC-08, TC-09).	**ACHIEVED**
Examination Operations & Repeats	ECA/ESA workflow segregation, 3-repeat attempt limits, bank voucher verification, and admission cards (TC-10, TC-11, TC-21).	**ACHIEVED**
Results & Formula Precision	Exact credit-weighted SGPA/CGPA algorithm according to Faculty Handbook Table 5.5.1 and Section 5.8 (TC-12).	**ACHIEVED**
Academic Progression Warnings	Automated detection of CGPA < 2.00 after Semester 3, barring unqualified Semester 5 registrations (TC-13, TC-18).	**ACHIEVED**
Welfare, Societies & Notifications	Directory of scholarships/societies and persistent real-time notifications with single and bulk delete (TC-15).	**ACHIEVED**
AI-Powered Process Guidance	Interactive natural language assistant grounded in faculty regulations via Gemini API with procedural search (TC-14).	**ACHIEVED**
Extensible Architecture	Modular Express architecture allowing new faculty workflows to be added with zero regression on existing modules.	**ACHIEVED**

8. Deployment and Setup Guide

8.1 Application Access

Live Production URL: https://seu-connect-plum.vercel.app

The live environment is configured with pre-seeded demonstration accounts for all institutional roles. Users can authenticate directly using the standard Institutional Sign In interface with sample passwords (`123456`): Student (`Student@gmail.com`), Lecturer (`Lecturer@gmail.com`), HOD (`HOD@gmail.com`), SAR (`sar@gmail.com`), Dean (`Dean@gmail.com`), Examination Officer (`Examination@gmail.com`), Dept MA (`DeptMA@gmail.com`), Faculty MA (`Fma@gmail.com`), and Administrator (`Admin@gmail.com`).

8.2 Local Installation and Execution

Prerequisites: Node.js (v18.0 or higher), npm (v9.0 or higher), Git, and active network access for MongoDB Atlas cloud connectivity.
Execution Commands:

```
# 1. Clone the repository and navigate into project root
git clone https://github.com/BinMushan/SEUconnect.git
cd SEUconnect

# 2. Start the Backend API Server (Port 5000)
cd Backend
npm install
npm start # Runs on http://localhost:5000

# 3. In a separate terminal, start the Frontend React Client (Port 5174)
cd ../Frontend/SEUconnect
npm install
npm run dev # Launches Vite development server on http://localhost:5174
```

8.3 Environment Configuration

All environment variables are managed securely via root and module `.env` files (never committed to public version control):

Variable Name	Scope	Functional Purpose in Platform
`MONGODB_URI`	Backend	MongoDB Atlas connection string with credentials and cluster endpoint.
`MONGODB_DB_NAME`	Backend	Target database name (`seuconnect`).
`JWT_SECRET`	Backend	High-entropy 256-bit cryptographic secret for signing and verifying tokens.
`PORT`	Backend	HTTP listen port for Express backend (default: `5000`).
`NODE_ENV`	Backend	Execution mode (`development` or `production`).
`GEMINI_API_KEY`	Backend	Google Gemini AI API key powering the RAG assistant (server-side only).
`VITE_API_URL`	Frontend	Base URL for API requests (`http://localhost:5000/api` locally, `/api` on Vercel).

8.4 Known Setup or Deployment Issues & Solutions

• **CORS Whitelisting:** When running frontend and backend on differing development ports (5174 and 5000), Express CORS middleware dynamically authorizes the client origin. On Vercel, client and API are unified under the same domain via `vercel.json` rewrites, bypassing cross-origin limits entirely.
• **Vercel Client-Side Routing:** React Router SPAs require URL rewrite rules (`/(.*) → /index.html`) to prevent 404 errors on browser page reloads. This was resolved via the committed root and frontend `vercel.json` configurations.
• **MongoDB Atlas IP Whitelisting:** Cloud Atlas deployments require Network Access whitelisting (`0.0.0.0/0`) to allow dynamic Vercel serverless IP ranges to connect.

9.1 Repository and Collaboration Workflow

The project was managed collaboratively on GitHub: https://github.com/BinMushan/SEUconnect. The team employed a feature-branching workflow (`Student-update-05`, `backend-update-1`, `Development`) merged into `main` via peer-reviewed Pull Requests (PRs #75 through #82). Commit histories document individual contributions: System Administration and Architecture by M.M.B. Mushan; Student Portal and SAR Desk by M.N.M. Afnan; HOD and Dean Desks by K. Jelaxsi; Lecturer Desk by J. Kamsa; and Examination Desk by M.R. Shahnas.

10. Limitations and Future Enhancements

10.1 Current Limitations

While SEUConnect provides full functional digitisation, three practical boundaries exist: (1) *Statutory Physical Document Verification:* University by-laws mandate that physical medical certificates and bank paying-in slips must still be verified against original paper copies prior to final Senate ratification; (2) *Sample Cohort Data:* The platform is currently populated with representative data for BICT cohorts (10 active students, 6 subjects); (3) *SMS Gateway Absences:* Urgent notices are delivered through the in-app notification center rather than cellular SMS broadcasts due to gateway costs.

10.2 Future Enhancements

The architectural modularity of SEUConnect enables straightforward integration of several planned future extensions:

• **Academic Advisor Portal:** Dedicated interface for faculty advisors to conduct formal load assessments and issue digital consent for credit ceiling deviations (Section 7.3).

• **Biometric / RFID Classroom Integration:** Hardware IoT integration with lecture hall card readers to log physical attendance directly into the Mongoose attendance collection.

• **LMS & External ERP Interoperability:** RESTful webhooks connecting SEUConnect with the university Moodle Virtual Learning Environment (VLE) for continuous assessment sync.

• **Automated SMS / Push Gateway:** Integration with telecommunication SMS gateways to broadcast urgent attendance warning alerts when a student drops below 82%.

• **Digital Certificate Cryptographic Verification:** Generating QR-coded, digitally signed exam admission cards and official GPA transcripts verifiable by third parties.

11. Conclusion

SEUConnect represents a transformative academic digitization initiative for the Faculty of Technology, South Eastern University of Sri Lanka. By replacing error-prone manual paper filing with a modern, reactive MERN-stack platform, the system effectively bridges the operational gap between undergraduates and faculty administrative authorities. The implementation faithfully adheres to the statutory provisions of the *Undergraduate Handbook 2023/2024*, strictly automating the 80% attendance gating threshold, 22-credit semester load ceiling, repeat attempt rules, multi-stage petition workflows, and credit-weighted GPA calculations. Through its seven tailored role dashboards and automated state-machine transitions, SEUConnect empowers students with real-time academic transparency while significantly reducing clerical latency for faculty officers. The project successfully demonstrates full-stack software engineering excellence, resilient cloud deployment, and collaborative teamwork, providing an extensible foundation ready for institutional adoption across SEUSL.

References

[1] South Eastern University of Sri Lanka, Faculty of Technology. (2023/2024). *Undergraduate Handbook 2023/2024*. Oluvil: SEUSL Press.

[2] Group C. (2026). *SEUConnect: Proposal for CA1 Group Assignment, SWT22022 Practical for Internet Application Development*. Faculty of Technology, SEUSL.

[3] Meta Platforms, Inc. (2025). *React 19 Documentation*. Retrieved from https://react.dev

[4] OpenJS Foundation. (2025). *Node.js v20 LTS Technical Documentation*. Retrieved from https://nodejs.org

[5] Express.js Foundation. (2025). *Express.js 5.0 Guide and API Reference*. Retrieved from https://expressjs.com

[6] MongoDB, Inc. (2025). *MongoDB Manual & Mongoose 9 ODM Documentation*. Retrieved from https://www.mongodb.com/docs/

[7] Internet Engineering Task Force (IETF). (2015). *RFC 7519: JSON Web Token (JWT) Standard*. Retrieved from https://datatracker.ietf.org/doc/html/rfc7519

[8] Google DeepMind. (2025). *Gemini 1.5 Flash API Documentation & Prompt Engineering Guide*. Retrieved from https://ai.google.dev/gemini-api/docs

[9] Vercel Inc. (2025). *Vercel Serverless Functions and SPA Deployment Specification*. Retrieved from https://vercel.com/docs

[10] Vite Foundation. (2025). *Vite: Next Generation Frontend Tooling*. Retrieved from https://vitejs.dev

Appendices

Appendix A – Key System Interface Highlights

• **Institutional Sign In:** Zero-scroll card featuring university branding, SVG credential inputs, institutional validation, and forgot-password recovery.
• **Student Academic Progress Trend:** Modern SVG curve charting historical semester GPAs against First Class (3.70) and Second Upper (3.30) benchmarks.
• **Department MA Desk:** Operational queue allowing Mrs. K. N. Fathima to audit 14-day statutory windows and approve subject medical excuses independently.
• **SAR Command Center:** Workflow command center provisioning examination voucher verification, admission generation, and ECA/ESA portal lock controls.

Appendix B – Core REST API Endpoint Reference Summary

Method	Endpoint Route	Functional Purpose & Action	Authorization Role
`POST`	`/api/auth/login`	Authenticates credentials; issues signed JWT bearer token	Public
`POST`	`/api/auth/register-student`	Provisions new student profile with generated password	**Admin only**
`POST`	`/api/auth/forgot-password`	Verifies student identity and executes password reset	Public / Student
`GET`	`/api/student/profile`	Fetches authenticated student details, batch, and GPA	Student
`GET`	`/api/student/attendance`	Returns module-wise attendance percentages & ESA eligibility	Student
`POST`	`/api/student/register`	Submits semester course enrollment with 22-cr limit check	Student
`GET`	`/api/student/results`	Retrieves transcript grades, SGPA, and cumulative CGPA	Student
`GET`	`/api/processes/inbox`	Fetches pending approval requests matching caller's role	**Staff / HOD / Dean / SAR**
`POST`	`/api/processes/request`	Initiates multi-stage petition (ECA, ESA, Medical, Exam Form)	Student
`PUT`	`/api/processes/:id/action`	Transitions petition stage (Approve, Return, Decline)	**Authorized Role**
`GET`	`/api/notifications`	Retrieves role-tailored notifications with unread counter	Authenticated
`DELETE`	`/api/notifications/:id`	Permanently deletes single notification from user feed	Authenticated
`DELETE`	`/api/notifications/clear-all`	Purges all active notifications for authenticated user	Authenticated
`GET`	`/api/health`	System health telemetry, uptime, and database status	Public

Appendix C – Technical State Machine & Schema Indexes

• **Compound Database Indexes:** `Students.createIndex({ regNo: 1 }, { unique: true })`, `ProcessRequests.createIndex({ trackingNumber: 1 }, { unique: true })`, `Attendance.createIndex({ student: 1, courseCode: 1 }, { unique: true })`, `Notifications.createIndex({ recipientRole: 1, createdAt: -1 })`.
• **Workflow States:** `SUBMITTED` → `LECTURER_REVIEW` → `HOD_ENDORSED` → `SAR_VERIFIED` → `EXAM_BRANCH_PROCESSED` → `DEAN_RATIFIED` → `SENATE_APPROVED`.